

AMC 10 Crash Course • Block 4 • Counting and Probability

Theory, worked examples, and independent sets. Solutions to the independent sets are at the end of the file.

Lesson 4.1. Sum and Product Rules, Permutations, Constraints

THEORY

Two rules solve almost everything. If a choice splits into non-overlapping cases, the counts **add** (sum rule). If a choice proceeds in steps and the number of options at each step does not depend on the previous ones, the counts **multiply** (product rule). The word “or” signals addition; “and then” signals multiplication.

The slot method. Draw empty positions and fill them one by one, starting with the **most restricted** slot. A three-digit number with distinct digits: first slot — 9 options (no zero), second — 9 (anything except the used digit), third — 8. Total $9 \cdot 9 \cdot 8 = 648$. Start with the wrong slot and the counts begin to depend on history — then split into cases.

Permutations of n distinct objects in a row: $n! = n \cdot (n - 1) \cdot \dots \cdot 1$. Arrangements of k out of n : $n \cdot (n - 1) \cdot \dots \cdot (n - k + 1)$ (k factors).

“Together” means glue. Two people must stand together: glue them into one block, permute the blocks, then multiply by the arrangements inside the block ($\times 2$ for a pair). **“Not together” means complement:** all arrangements minus those where they are together. Counting “not together” directly is slower and riskier.

Error control. The main traps of this lesson: a forgotten zero in the leading digit, slots filled in a bad order, and double counting. Symmetry helps too: when the conditions treat objects symmetrically (say, equally many even and odd digits), symmetric cases must give equal counts. Always test yourself on a tiny example: if the formula is right for 2–3 objects, where everything can be listed by hand, you can trust it.

WORKED EXAMPLES

EXAMPLE 1 • SLOT METHOD

How many three-digit numbers have all digits distinct?

→ Slots left to right: 9 (first digit is not zero) $\cdot$ 9 (anything except the first digit — zero is now allowed) $\cdot$ 8 = **648**. The trap is writing $9 \cdot 10 \cdot 10$ and forgetting distinctness, or $10 \cdot 9 \cdot 8$ and letting zero lead.

EXAMPLE 2 • “EVEN” AND CASES

How many even three-digit numbers have all digits distinct?

→ Start with the most restricted slot — the last one. It drags zero into the picture, so two cases. Last digit 0: $9 \cdot 8 = 72$. Last digit in $\{2, 4, 6, 8\}$: $4 \cdot 8$ (first digit: not zero and not the last digit) $\cdot 8 = 256$. Total $72 + 256 =$ **328**. The trap is writing $5 \cdot 8 \cdot 8$ in one stroke: zero and 2–8 impose different restrictions on the first digit.

EXAMPLE 3 • GLUING

Five people line up for a photo. Ann and Ben want to stand next to each other. In how many ways can everyone be arranged?

→ Glue Ann and Ben into one block: 4 objects, $4! = 24$ arrangements. Inside the block there are two orders: AB and BA. Total $24 \cdot 2 =$ **48**. The classic mistake is forgetting the factor of 2 for the order inside the glued block.

EXAMPLE 4 • “NOT ADJACENT” VIA COMPLEMENT

Six books are placed on a shelf. The two dictionary volumes must not stand next to each other. How many arrangements are there?

→ All arrangements minus “dictionaries together”: $6! - 2 \cdot 5! = 720 - 240 = 480$. Counting “not adjacent” directly is a mess of positions; the complement is one line. Build the reflex: a “no-adjacency” ban is almost always counted by subtracting the glued case.

INDEPENDENT SET

1. A cafe offers 4 soups, 5 mains, and 3 drinks. In how many ways can you build a lunch of one soup, one main, and one drink?
2. How many four-letter codes with no repeated letters can be made from the letters A, B, C, D, E?
3. Five people line up in a row. Ann must stand at an end (either one). How many arrangements are there?
4. How many three-digit numbers have all digits odd?
5. How many three-digit numbers with all digits distinct are divisible by 5?
6. Four boys and three girls line up so that all three girls stand together. How many arrangements are there?
7. Five people line up in a row. Pete and Vic had a fight and refuse to stand next to each other. How many arrangements are there?
8. Seven people line up in a row. In how many arrangements does Ann stand somewhere to the left of Ben (not necessarily adjacent)?
9. ★ Five-digit numbers are formed from the digits 1, 2, 3, 4, 5, each digit used exactly once. How many of them are divisible by 4?

Lesson 4.2. Combinations: Team Selection, $C(n, k)$, Stars and Bars

THEORY

Order matters or not — the central question of counting. Choosing a captain and a goalkeeper — order matters, that is an arrangement. Choosing two monitors — order does not matter, that is a combination. The number of ways to choose k objects out of n without order: $C(n, k)$ = the arrangements divided by $k!$ — every group was counted $k!$ times, and we pay the debt back by dividing.

$$C(n, k) = \frac{n \cdot (n-1) \cdot \dots \cdot (n-k+1)}{k!} \cdot \text{Properties: } C(n, k) = C(n, n-k); C(n, 0) = C(n, n) = 1.$$

Group choices multiply. A team of 2 boys and 2 girls: $C(5, 2) \cdot C(6, 2)$ — the choices are independent. The symmetry $C(n, k) = C(n, n-k)$ saves arithmetic: compute $C(10, 8)$ as $C(10, 2) = 45$. Useful images: handshakes among n people — $C(n, 2)$; diagonals of an n -gon — $C(n, 2) - n$ (pairs of vertices minus the sides).

Stars and bars. In how many ways can n identical candies be given to k children? Lay the candies in a row and insert $k-1$ bars. If everyone gets at least one, the bars go into the gaps between candies: $C(n-1, k-1)$ ways. If zero is allowed, first hand each child a “virtual” candy: $C(n+k-1, k-1)$.

The trap of this lesson is double counting. If you pick “one special member first, then the rest”, check: are you counting the same group several times with different “specials”? The standard cure: count directly with combinations, or divide by the number of repeats.

WORKED EXAMPLES

EXAMPLE 1 • ORDER DOES NOT MATTER

Three students are chosen for a team (no roles) out of 10. In how many ways?

→ Order does not matter: $C(10, 3) = 10 \cdot 9 \cdot 8 / 3! = 720/6 = \mathbf{120}$. The answer 720 is the classic order-versus-choice trap: 720 counts triples with roles, so every team is counted 6 times.

EXAMPLE 2 • CHOOSING BY GROUPS

A club has 5 boys and 6 girls. In how many ways can a delegation of 2 boys and 2 girls be chosen?

→ The choices are independent and multiply: $C(5, 2) \cdot C(6, 2) = 10 \cdot 15 = \mathbf{150}$. The trap is computing $C(11, 4)$: that would allow delegations with three boys.

EXAMPLE 3 • DIAGONALS

How many diagonals does a convex decagon have?

→ Every pair of vertices gives a segment: $C(10, 2) = 45$. Of these, 10 are sides, not diagonals: $45 - 10 = \mathbf{35}$. Forgetting to subtract the sides is the routine error; the second one is computing $10 \cdot 7$ and not dividing by 2 (each diagonal has two endpoints).

EXAMPLE 4 • STARS AND BARS

How many solutions in positive integers does $x + y + z = 10$ have?

→ Ten stars in a row, 9 gaps, place 2 bars: $C(9, 2) = \mathbf{36}$. The trap is mixing up the modes: for nonnegative integers it would be $C(12, 2) = 66$. Read the problem twice: “positive” means everyone gets at least one.

INDEPENDENT SET

1. In how many ways can 3 books be chosen out of 8?
2. Twelve people at a meeting each shake hands with everyone else exactly once. How many handshakes are there?
3. A committee of 3 men and 2 women is chosen from 6 men and 4 women. In how many ways?
4. How many diagonals does a convex 12-gon have?
5. In a tournament every player played every other exactly once, 45 games in total. How many players were there?
6. Five books are chosen out of 9, but the two encyclopedia volumes cannot both be taken. In how many ways?
7. In how many ways can 12 identical candies be given to four children so that each gets at least one?
8. A grid is made of 5×4 unit cells (6 vertical and 5 horizontal lines). How many rectangles in total can be seen in this grid?
9. ★ In how many ways can 6 people be split into 3 pairs for table tennis?

Lesson 4.3. Casework, Complementary Counting, Inclusion-Exclusion

THEORY

Casework is honest labor. When no single formula applies, cut the problem into non-overlapping cases and add. Two requirements for the cut: the cases must not overlap and together must cover everything. Choose the cutting parameter so that there are few cases: cutting “by the last digit” usually beats cutting “by the first”.

“At least one” is the signal for the complement. Counting “at least one” directly means adding up a dozen cases and almost surely double-counting something. Flip it: (at least one) = (total) – (none). “None” is almost always a single product. This is the most profitable reflex in all of counting.

$$|A \cup B| = |A| + |B| - |A \cap B|. \text{ For three sets: } |A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

Inclusion-exclusion. Add two sets and the intersection is counted twice — subtract it once. With three sets: add the singles, subtract the pairwise intersections, add the triple back. For divisibility: the count of numbers from 1 to N divisible by d is exactly $\lfloor N/d \rfloor$; the intersection “divisible by 3 and 5” means divisible by 15 — the LCM, not the product, when the numbers are not coprime.

Combine the tools. The ban “ A not first” and “ B not last” is complement plus inclusion-exclusion: total – (bad by A) – (bad by B) + (bad by both). The final addition is mandatory: what was subtracted twice must be given back.

WORKED EXAMPLES

EXAMPLE 1 • COMPLEMENT

How many three-digit numbers contain at least one digit 7?

→ “At least one” — flip it. No sevens at all: $8 \cdot 9 \cdot 9 = 648$. Total three-digit numbers: 900. Answer: $900 - 648 = \mathbf{252}$. A direct count (7 in the first place, in the second, in the third...) drowns in double-counting numbers like 774.

EXAMPLE 2 • TWO SETS

How many integers from 1 to 100 are divisible by 3 or by 5?

→ Divisible by 3: 33; by 5: 20; by 15: 6. By the formula: $33 + 20 - 6 = \mathbf{47}$. Skipping the subtraction is the main failure: 15, 30, ... are counted in both lists.

EXAMPLE 3 • CASEWORK

How many three-digit numbers have digit sum 5?

→ Casework on the first digit $a = 1, 2, 3, 4, 5$: the remainder $5 - a$ splits into two digits in 5, 4, 3, 2, 1 ways respectively. Sum: $5 + 4 + 3 + 2 + 1 = \mathbf{15}$. Check the cut: the cases do not overlap and cover everything. (A connoisseur will spot stars and bars here — same answer.)

EXAMPLE 4 • THREE SETS

Of 100 students, 60 take math, 50 physics, 40 chemistry; 30 take math and physics, 25 math and chemistry, 20 physics and chemistry, and 10 take all three. How many students take none of the subjects?

→ The union: $60 + 50 + 40 - 30 - 25 - 20 + 10 = 85$. Taking none: $100 - 85 = \mathbf{15}$. The trap is forgetting to add the triple intersection back: students taking all three were added three times and subtracted three times, so they must be restored.

INDEPENDENT SET

1. How many two-digit numbers have digit sum 9?
2. A coin is tossed 5 times. How many outcome sequences contain at least one head?
3. How many integers from 1 to 60 are divisible by 4 or by 6?
4. A class has 25 students: 17 like math, 15 like English, and every student likes at least one of the two. How many like both?
5. How many four-digit numbers contain at least one digit 0?
6. A committee of three with at least one woman is chosen from 5 men and 4 women. In how many ways?
7. How many integers from 1 to 1000 are divisible by at least one of 2, 3, 5?
8. How many permutations of the letters A, B, C, D, E have A not in the first position and B not in the last?
9. ★ How many subsets of $\{1, 2, \dots, 10\}$ contain at least one even AND at least one odd number?

Lesson 4.4. Probability via Counting: Dice, Coins, Cards

THEORY

Probability = a fraction made of two counts. When all outcomes are equally likely, $P = (\text{favorable}) / (\text{total})$. All the difficulty lives in the counting of numerator and denominator, so lessons 4.1–4.3 keep working here. The iron rule: count numerator and denominator **in the same model** — either both ordered, or both unordered.

$$P(A) = \frac{\text{number of favorable outcomes}}{\text{number of all outcomes}} \cdot P(\text{at least one}) = 1 - P(\text{none}).$$

The standard playgrounds. Two dice: 36 ordered pairs — keep them ordered: the sum 8 comes from five pairs (2, 6), (3, 5), (4, 4), (5, 3), (6, 2), not three. Coins: n tosses give 2^n sequences, and exactly k heads happens in $C(n, k)$ of them. A standard 52-card deck: 4 suits of 13 ranks; two cards without replacement are best handled step by step: $P(\text{both hearts}) = 13/52 \cdot 12/51$ — the second fraction already “knows” about the first card.

Geometric probability. When there are infinitely many outcomes (a point on a segment, a pair of numbers), probability = (length or area of the favorable region) / (length or area of the whole). Draw the picture: the square of possibilities and the condition’s region inside it.

The answer is a fraction in lowest terms. On the AMC a probability is almost always a fraction, and a wrongly reduced fraction sits among the choices. Sanity checks are mandatory: “at least one” is at least as likely as “exactly one”, and every probability lives between 0 and 1.

WORKED EXAMPLES

EXAMPLE 1 • TWO DICE

Two dice are rolled. What is the probability that the sum is 8?

→ There are 36 ordered pairs. Sum 8: (2, 6), (3, 5), (4, 4), (5, 3), (6, 2) — five pairs. $P = \frac{5}{36}$. The trap is treating (2, 6) and (6, 2) as one outcome: then the outcomes stop being equally likely, since (4, 4) occurs half as often as “2 and 6 in some order”.

EXAMPLE 2 • COINS

A coin is tossed three times. What is the probability of exactly two heads?

→ There are $2^3 = 8$ sequences; exactly two heads — $C(3, 2) = 3$ of them (HHT, HTH, THH). $P = \frac{3}{8}$. The error “four outcomes: 0, 1, 2, 3 heads” forgets that those four outcomes are not equally likely.

EXAMPLE 3 • CARDS

Two cards are drawn without replacement from a standard 52-card deck. What is the probability that both are hearts?

→ Step by step: $\frac{13}{52} \cdot \frac{12}{51} = \frac{1}{4} \cdot \frac{4}{17} = \frac{1}{17}$. The second fraction is 12/51, not 13/52: one heart is already gone. The same answer via combinations: $C(13, 2)/C(52, 2)$ — the unordered model in the numerator AND the denominator.

EXAMPLE 4 • GEOMETRIC

Numbers x and y are chosen independently and uniformly from $[0, 1]$. What is the probability that $x + y < 1/2$?

→ The pair (x, y) is a point of the unit square of area 1. The condition cuts off a right triangle with legs 1/2: area $(1/2) \cdot (1/2) \cdot (1/2) = \frac{1}{8}$. Answer: $\frac{1}{8}$. Enumerating outcomes is impossible here in principle — only area works.

INDEPENDENT SET

1. A die is rolled once. What is the probability of rolling a prime number?
2. A coin is tossed 4 times. What is the probability of exactly two heads?

3. Two dice are rolled. What is the probability that the sum is 7?
4. One card is drawn from a standard 52-card deck. What is the probability that it is a face card (jack, queen, or king)?
5. Two dice are rolled. What is the probability that at least one six appears?
6. A bag holds 5 red and 3 blue marbles. Two are drawn at random. What is the probability that they are different colors?
7. Two cards are drawn without replacement from a standard 52-card deck. What is the probability that both are aces?
8. Numbers x and y are chosen independently and uniformly from $[0, 3]$. What is the probability that $x > y + 1$?
9. ★ Three standard dice are rolled. What is the probability that one of the values equals the sum of the other two?

Lesson 4.5. Lattice Paths, Recursive Counting, Expected Value

THEORY

Lattice paths. A path from $(0, 0)$ to (m, n) by unit steps right and up is a word of m letters R and n letters U. There are as many paths as letter arrangements: $C(m + n, m)$. A path “through point P ” is a product: (to P) $\cdot$ (after P). A path “avoiding P ” is a complement: all minus through P .

Paths from $(0, 0)$ to (m, n) : $C(m + n, m)$. Stairs by 1 or 2 steps: $f(n) = f(n - 1) + f(n - 2)$. $E = \sum (\text{value} \cdot \text{probability})$.

Recursive counting. When no direct formula is in sight, ask: how does the object end? A climb of n steps ends with a 1-step (preceded by any climb of $n - 1$) or a 2-step (a climb of $n - 2$): $f(n) = f(n - 1) + f(n - 2)$ — the Fibonacci numbers. The same skeleton counts domino tilings of a $2 \times n$ strip and binary words with no two adjacent ones. Compute the first values by hand ($f(1) = 1, f(2) = 2$ for the stairs) and unroll the table — a wrong start ruins everything.

Expected value — the minimum. The expected value is the average payoff over many repetitions: the sum of value $\times$ probability. For a die: $(1 + 2 + \dots + 6)/6 = 3.5$. The expectation of a sum equals the sum of expectations — no independence required.

Test strategy. Small grids are faster to fill with numbers right on the picture (each node = left neighbor + bottom neighbor) than to attack by formula. It is the same recursive principle, and it is immune to obstacle errors: a forbidden node simply gets a zero.

WORKED EXAMPLES

EXAMPLE 1 • PATHS

How many paths lead from $(0, 0)$ to $(4, 3)$ by unit steps right and up?

→ Every path is a word of 4 R’s and 3 U’s: choose the places for U among 7 steps, $C(7, 3) = \mathbf{35}$. The order-versus-choice trap in reverse: nothing needs an extra factor — the word already determines the path uniquely.

EXAMPLE 2 • AVOIDING A POINT

How many paths from $(0, 0)$ to $(4, 4)$ do not pass through $(2, 2)$?

→ Complement. Total: $C(8, 4) = 70$. Through $(2, 2)$: $C(4, 2) \cdot C(4, 2) = 36$ — the product of “before” and “after”. Answer: $70 - 36 = \mathbf{34}$. Counting “avoiding” directly means chasing routes along the edges and losing some.

EXAMPLE 3 · STAIRS

A staircase of 10 steps is climbed in moves of 1 or 2 steps. How many ways are there to climb it?

→ The last move: a 1-step (preceded by a climb of 9) or a 2-step (a climb of 8): $f(10) = f(9) + f(8)$. The table from $f(1) = 1, f(2) = 2$: 1, 2, 3, 5, 8, 13, 21, 34, 55, **89**. The main error is the false start $f(2) = 1$: there are TWO ways up two steps (1+1 and 2).

EXAMPLE 4 · EXPECTED VALUE

A die is rolled, and you are paid as many dollars as the number shown. What is the expected payoff?

→ Each value has probability $1/6$: $E = (1 + 2 + 3 + 4 + 5 + 6)/6 = 21/6 = \frac{7}{2}$ dollars. The expectation need not be a possible outcome: 3.5 never shows, yet that is exactly the average payoff per roll.

INDEPENDENT SET

1. How many paths lead from (0, 0) to (3, 3) by steps right and up?
2. How many paths lead from (0, 0) to (5, 2) by steps right and up?
3. In how many ways can a staircase of 8 steps be climbed in moves of 1 or 2?
4. In how many ways can a 2×9 strip be tiled with 1×2 dominoes?
5. How many paths from (0, 0) to (4, 4) pass through the point (1, 1)?
6. A die is rolled, and the payoff is the square of the number shown (in dollars). What is the expected payoff?
7. How many paths from (0, 0) to (3, 3) avoid the point (1, 1)?
8. How many words of length 10 over the letters A and B contain no two adjacent B's?
9. ★ How many paths from (0, 0) to (5, 5) by unit steps right and up avoid both (1, 1) and (4, 4)?

Solutions to the independent sets

LESSON 4.1

1. $4 \cdot 5 \cdot 3 = 60$.

2. $5 \cdot 4 \cdot 3 \cdot 2 = 120$.

3. 2 ends for Ann, the rest freely: $2 \cdot 4! = 48$.

4. $5 \cdot 5 \cdot 5 = 125$. Zero is even, so the leading-digit restriction costs nothing here.

5. Ending 0: $9 \cdot 8 = 72$. Ending 5: 8 (first digit: not zero and not 5) $\cdot 8 = 64$. Total **136**.

6. The girls' block + 4 boys = 5 objects: $5! \cdot 3! = 120 \cdot 6 = 720$.

7. $5! - 2 \cdot 4! = 120 - 48 = 72$.

8. In exactly half of all $7!$ arrangements Ann is to the left: $5040/2 = 2520$. No casework — pure symmetry.

★ The valid endings: 12, 24, 32, 52 — four of them. The first three digits are any arrangement of the rest: $3! = 6$. Total: $4 \cdot 6 = 24$.

LESSON 4.2

1. $C(8, 3) = 8 \cdot 7 \cdot 6 / 6 = 56$.

2. $C(12, 2) = 66$.

3. $C(6, 3) \cdot C(4, 2) = 20 \cdot 6 = 120$.

4. $C(12, 2) - 12 = 66 - 12 = 54$.

5. $n(n-1)/2 = 45$, so $n(n-1) = 90$ and $n = 10$.

6. $C(9, 5) - C(7, 3) = 126 - 35 = 91$ (if both volumes are in, we pick 3 more out of 7).

7. $C(11, 3) = 165$.

8. $C(6, 2) \cdot C(5, 2) = 15 \cdot 10 = 150$. No size-by-size enumeration: two pairs of lines determine the rectangle uniquely.

★ $15 \cdot 6 \cdot 1 = 90$ treats the pairs as “first, second, third” — every split is counted $3! = 6$ times. Answer: $90/6 = 15$.
Overcounting in its purest form: divide by the permutations of indistinguishable groups.

LESSON 4.3

1. 18, 27, ..., 90: one for each first digit, **9** in total.

2. $2^5 - 1 = 31$: only one sequence has no heads — all tails.

3. $15 + 10 - 5 = 20$.

4. $17 + 15 - x = 25$, so $x = 7$.

5. $9000 - 9^4 = 9000 - 6561 = 2439$.

6. $C(9, 3) - C(5, 3) = 84 - 10 = 74$.

7. $500 + 333 + 200 - 166 - 100 - 66 + 33 = 734$.

8. $120 - 24 - 24 + 6 = 78$. The +6 term is mandatory: permutations with both offenses were subtracted twice.

★ In total $2^{10} = 1024$. With no evens: $2^5 = 32$; with no odds: 32; the empty set was subtracted twice. Answer: $1024 - 32 - 32 + 1 = 961$.

LESSON 4.4

1. 3 out of 6: $\frac{1}{2}$. One is not prime!

2. $6/16 = \frac{3}{8}$.

3. Six pairs out of 36: $\frac{1}{6}$. Seven is the most likely sum on two dice.

4. $12/52 = \frac{3}{13}$.

5. $1 - (5/6)^2 = 1 - 25/36 = \frac{11}{36}$.

6. $5 \cdot 3 = 15$ mixed pairs out of $C(8, 2) = 28$: $\frac{15}{28}$.

7. $\frac{4}{52} \cdot \frac{3}{51} = \frac{1}{221}$.

8. The region $x - y > 1$ is a right triangle with legs 2, area 2. Probability $2/9$: $\frac{2}{9}$.

★ There are $5 + 4 + 3 + 2 + 1 = 15$ ordered pairs (x, y) with $x + y \leq 6$; the “sum” die can sit in any of 3 positions, and no outcome is counted twice (two dice cannot both equal the sum of the others). That is 45 outcomes out of 216: probability $45/216 = 5/24$.

LESSON 4.5

1. $C(6, 3) = 20$.

2. $C(7, 2) = 21$.

3. 1, 2, 3, 5, 8, 13, 21, **34**.

4. The same recurrence: $t(n) = t(n-1) + t(n-2)$, start 1, 2: we get **55**.

5. $C(2, 1) \cdot C(6, 3) = 2 \cdot 20 = 40$.

6. $(1 + 4 + 9 + 16 + 25 + 36)/6 = \frac{91}{6}$.

7. $20 - C(2, 1) \cdot C(4, 2) = 20 - 12 = 8$.

8. $g(n) = g(n-1) + g(n-2)$, start $g(1) = 2, g(2) = 3$: 2, 3, 5, 8, 13, 21, 34, 55, 89, **144**.

★ Total: $C(10, 5) = 252$. Through (1, 1): $2 \cdot C(8, 4) = 140$; through (4, 4): $C(8, 4) \cdot 2 = 140$; through both: $2 \cdot C(6, 3) \cdot 2 = 80$. Answer: $252 - 140 - 140 + 80 = 52$. Lesson 4.3 works on the grid too: give back what was subtracted twice.